Tumor primary site as a prognostic factor for Merkel cell carcinoma disease-specific death

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Background: Merkel cell carcinoma (MCC) primary site has not been fully investigated as a potential prognostic factor.

Objective: To determine the incidence by tumor primary site of death due to MCC.

Methods: We undertook a retrospective analysis of the Survival, Epidemiology, and End Results database. MCC patients treated between 1973 and 2016 were grouped by tumor primary site and a competing risks analysis was performed to test the impact of primary site on disease-specific death. Cumulative incidence of Merkel cell carcinoma-specific mortality (CMMI) at 5 years was estimated for each primary site.

Results: Of 9407 MCC patients identified, 6305 (67.0%) had localized disease, 2397 (25.5%) had regional metastasis, and 705 (7.5%) had distant metastasis. Tumor primary site was predictive of CMMI and varied by stage at diagnosis. Tumors involving the scalp/neck carried the highest CMMI among localized MCC (26.0%). Tumors involving the lip had the highest CMMI among MCC with regional metastasis (56.7%) and distant metastasis (82.1%).

Limitations: Tumor size data were missing for a large proportion of patients, precluding stratification by stage according to current American Joint Committee on Cancer guidelines.

Conclusions: Probability of MCC disease-specific death varies by primary site. The primary site of the tumor may be useful as a prognostic indicator for MCC. (J Am Acad Dermatol 2021;85:1259-66.)

Key words: Merkel cell carcinoma; Merkel cell carcinoma survival; tumor primary site; tumor site; unknown primary site.

INTRODUCTION

Merkel cell carcinoma (MCC) is a rare, aggressive neuroendocrine malignancy typically occurring in older white patients and often located on the head and neck.[1] At diagnosis, up to 40% of patients will have regional and 8% will have distant lymph node metastases.[2-6] Recurrence is reported in about 22% of cases[5] and 5-year survival ranges from 13% to 65%.[3,7]

In 12% to 15% of cases, MCC will present in 1 or more lymph nodes without an evident cutaneous primary site.[6,8,9] The prognosis of MCC of unknown primary site compared to MCC with known primary

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site has been well studied, with evidence suggesting that an unknown primary site is associated with a better prognosis.[5,6,10,11] Although few studies have investigated the prognostic utility of tumor primary site beyond unknown primaries, 1 recently published study identified MCC in the head/neck region to have a worse overall survival than tumors in the extremities.[12] However, the impact of tumor primary site on disease-specific death has yet to be elucidated. As such, our objective was to compare disease-specific death due to MCC for various cutaneous primary sites using a national population-based cancer registry.

CAPSULE SUMMARY

- Disease-specific mortality rates for Merkel cell carcinoma vary both by broad anatomic region of tumor development and by granular sites of tumor development within a given region.
- Clinicians may consider tumor primary site as an additional prognostic factor for Merkel cell carcinoma.

METHODS

Study design

We retrospectively reviewed the Surveillance, Epidemiology, and End Results (SEER) database of 18 registries, identifying patients diagnosed with MCC between 1973 and 2016. SEER is a program of the National Cancer Institute, which collects data on the incidence of cancer from population-based cancer registries in 18 geographic regions across the United States.[13-15]

Patients with a pathology-confirmed diagnosis of MCC were included in this study. Patients were excluded if the tumor primary site was an internal visceral site (eg, lung, esophagus) or was listed as "unknown" but lacked designation as "MCC of unknown primary." Patients with a diagnosis of both MCC and small cell carcinoma were excluded due to similar histology and concerns for diagnostic accuracy.[16] For survival analysis only, we further excluded patients diagnosed at autopsy or reported by death certificate only, patients without survival data, patients with unknown or unclear cause of death, and patients who did not receive surgical intervention.[17] The flowchart in Supplemental Fig 1 (available via Mendeley at https://data.mendeley.com/datasets/k79w7xwfss/1) details exclusion criteria. This study was deemed exempt from Institutional Review Board review because publicly available, deidentified data were used.

Methods of measurement

Data on patient demographics (age at diagnosis, sex, and race), tumor characteristics (stage at diagnosis and tumor primary site), treatment modalities (chemotherapy or radiation), and survival (vital status, cause of death, and months of survival after diagnosis) were extracted from SEER. Patients were grouped by tumor primary site (ear, eyelid, lip, other parts of face, lower limb, upper limb, scalp/neck, trunk, or unknown primary) and stage at diagnosis (localized disease, regional metastasis, or distant metastasis).

"Distant metastasis" included metastatic disease extending to distant nodal basins or visceral sites. "Regional metastasis" included metastatic disease limited to regional nodal basins or in-transit metastasis. "Localized disease" included patients without metastatic disease. Our primary outcome was disease-specific death, measured by the cumulative Merkel cell carcinoma-specific mortality incidence (CMMI) for each tumor primary site.

Data analysis

Analyses were performed using SAS 9.4 (SAS Institute, Inc) and R, version 4.0.4 (R Foundation). Median overall survival with interquartile ranges (IQRs) was calculated for the sample population using the Kaplan-Meier estimate. We established a competing risks framework, in which MCC disease-specific death was the event of interest and death from any other cause was the competing risk. Fine and Gray's proportional subdistribution hazards regression was used to model CMMI for each tumor primary site (Supplemental Methods; available via Mendeley at https://data.mendeley.com/datasets/k79w7xwfss/1).[18] We assessed for first-degree interaction between tumor primary site and the stage at diagnosis and subsequently included this interaction term in the model. The final regression model was adjusted for age, sex, and receipt of adjuvant treatment. Race was not included for adjustment, as it was found to be neither a confounder nor an effect modifier in our model. Chemotherapy and radiation were included in the model, despite not confounding the effect of tumor primary site on MCC disease-specific death (Supplemental Table 1). To evaluate for accuracy of the adjusted model, we performed k-fold cross-validation and produced a calibration plot that graphically depicted the predicted against the observed probabilities of mortality.[19,20] A sensitivity analysis was performed to determine whether results remained consistent across the study period, with

Abbreviations used:

CMMI:	cumulative Merkel cell carcinoma-specific mortality incidence
IQR:	interquartile range
MCC:	Merkel cell carcinoma
SEER:	Surveillance, Epidemiology, and End Results
sHR:	subdistribution hazard ratio

stratification into 3 eras (1973-2000, 2001-2010, 2011-2016) allowing for analysis of differences in model estimates.

Results are reported as subdistribution hazard ratios (sHRs) with 95% confidence intervals (95% confidence intervals) and $P < .05$ was considered significant. Based on the Fine and Gray model, we created 3 graphs of CMMI versus time, overlaying all tumor primary sites, 1 for each localized disease, regional metastasis, and distant metastasis. We also estimated the CMMI at 5 years for each tumor primary site, given each stage at diagnosis.

RESULTS

We identified 9407 patients diagnosed with MCC from 1973 to 2016. Patient demographics and tumor characteristics are presented in Table I. At diagnosis, 6305 (67.0%) patients had localized disease, 2397 (25.5%) had regional metastasis, and 705 (7.5%) had distant metastasis. Overall, tumors occurred most often on the face (26.4%), not including the ear, eyelid, and lip, followed by the upper limb (24.0%), which was the most common primary site not part of the head/neck.

MCC of unknown primary site occurred most often for regional and distant metastatic disease stages (23.0% and 26.1%, respectively). After excluding patients without survival data and those who did not receive surgery, we conducted survival and competing risk analyses on 8190 patients. Death due to MCC occurred in 2015 (24.6%) patients and death due to other causes occurred in 2882 (35.2%), while 3293 (40.2%) were alive at last follow up. The median survival for localized tumors was 61 months (IQR, 21-153), compared to 35 months (IQR, 13-137) for regional metastatic tumors and 11 months (IQR, 6-27) for distant metastatic tumors ($P < .0001$).

Initial comparison of tumor primary sites by their unadjusted cumulative incidence curves revealed an overall significant difference in CMMI between sites. Multivariable analysis using Fine and Gray's subdistribution hazards regression revealed that tumor primary site had a significant effect on CMMI ($P < .0001$; Table II). At the localized disease stage, scalp/neck and truncal tumors had the greatest impact on mortality, followed by tumors of the lip, ear, lower limb, and other parts of face. Interaction testing revealed that the effect of tumor primary site on the sHR is modified by stage at diagnosis (χ^2 for joint test of interaction, 35.4; $P < .0021$), with the order of tumor primary site sHRs changing with increasing stage of disease (Fig 1). At both the regional and distant metastatic stages, sHR for all tumor sites except the lip decreased relative to the upper limb.

Fig 2 demonstrates the CMMI modeled from the subdistribution hazards. MCCs of the lip and upper limb demonstrate a disproportionate increase in CMMI with increasing stage compared to other tumor primary sites. MCCs of the eyelid demonstrate a disproportionately lower increase in CMMI with increasing stage.

The corresponding 5-year probability of mortality was highest for scalp/neck (26.0%) and truncal (25.9%) tumors for localized MCC, but highest for lip tumors for MCC with regional (56.7%) or distant (82.1%) metastasis (Table III). Eyelid tumors had the lowest 5-year probability of mortality across all disease stages (14.3% for localized MCC, 34.8% for MCC with regional metastasis, and 39.3% for MCC with distant metastasis). In comparison, MCC of unknown primary had 5-year probabilities of mortality of 31.9% (regional metastasis) and 58.9% (distant metastasis). K-fold cross-validation demonstrated strong agreement between predicted and observed mortality probabilities (Supplemental Fig 2). Sensitivity analysis stratified by era produced consistent estimates for the effect of tumor primary site on CMMI, although estimated mortality was higher for all tumor sites in 1973-2000 than in 2001-2010 or 2011-2016 (Supplemental Fig 3).

Discussion

We performed a retrospective analysis of SEER and found that the primary site of the MCC tumor impacts the incidence of disease-specific death and that this effect is modified by stage at diagnosis. Estimating CMMI (representing probability of death due to MCC) for each tumor primary site after 5 years of disease revealed that several primary sites carry greater a probability of disease-specific death over time. Specifically, scalp/neck and truncal tumors had highest CMMI with localized disease, but lip tumors had highest CMMI with both regional metastatic and distant metastatic disease.

Although MCC of the upper limb had more favorable outcomes than those of other primary sites with localized disease, these tumors appear to have a relatively greater incidence of mortality once metastasized. Differences in disease-specific death remain

Table I. Characteristics of patients with primary Merkel cell carcinoma (1973-2016)

	Overall (n = 9407)	Localized disease (n = 6305)	Regional metastasis (n = 2397)	Distant metastasis (n = 705)
Characteristics	n (%)	n (%)	n (%)	n (%)
Age at diagnosis, median (range)	77 (11-105)	78 (11-105)	74 (30-99)	76 (25-102)
Sex				
Male	5867 (62.4)	3770 (59.8)	1616 (67.4)	481 (68.2)
Female	3540 (37.6)	2535 (40.2)	781 (32.6)	224 (31.8)
Race				
White	8985 (95.3)	6008 (95.3)	2294 (95.7)	663 (94.0)
Black	114 (1.2)	66 (1.0)	33 (1.4)	15 (2.1)
Other	259 (2.8)	167 (2.7)	64 (2.7)	26 (3.8)
Unknown	69 (0.7)	64 (1.0)	4 (0.2)	1 (0.1)
Tumor primary site				
Ear	330 (3.5)	228 (3.6)	77 (3.2)	25 (3.5)
Eyelid	195 (2.1)	138 (2.2)	23 (1.0)	34 (4.8)
Lip	223 (2.4)	161 (2.5)	44 (1.8)	18 (2.6)
Face (other)	2480 (26.4)	1947 (30.9)	434 (18.1)	99 (14.0)
Lower limb	1359 (14.4)	935 (14.8)	344 (14.3)	80 (11.4)
Upper limb	2260 (24.0)	1714 (27.2)	453 (18.9)	93 (13.2)
Scalp/neck	895 (9.5)	618 (9.8)	198 (8.3)	79 (11.2)
Trunk	929 (9.9)	564 (9.0)	272 (11.4)	93 (13.2)
Unknown primary*	736 (7.8)	-	552 (23.0)	184 (26.1)

n, Number.
*Unknown primary cannot be defined as localized disease.

Table II. Subdistribution hazards model for disease-specific death in patients with primary Merkel cell carcinoma (1973-2016)

Characteristics		Subdistribution HR (95% CI)		*P* value
Age		1.02 (1.01-1.02)		<.0001
Sex				
Male		1.37 (1.35-1.50)		<.0001
Female		Reference		-
Chemotherapy				
Yes		1.74 (1.54-1.96)		<.0001
No		Reference		
Radiation				
Yes		0.93 (0.85-1.02)		.13
No		Reference		
	Localized disease	Regional metastasis	Distant metastasis	
Tumor primary site				
Scalp/neck	1.89 (1.50-2.36)	1.25 (0.94-1.65)	0.99 (0.60-1.63)	<.0001
Trunk	1.87 (1.49-2.34)	1.04 (0.81-1.33)	1.13 (0.73-1.75)	<.0001
Lip	1.63 (1.12-2.38)	1.44 (0.85-2.44)	1.66 (0.85-3.23)	.011
Ear	1.54 (1.10-2.17)	0.76 (0.48-1.21)	0.88 (0.44-1.76)	.011
Lower limb	1.51 (1.23-1.86)	0.94 (0.74-1.19)	0.95 (0.58-1.55)	<.0001
Face (other)	1.28 (1.07-1.53)	0.96 (0.76-1.20)	0.97 (0.64-1.48)	.008
Upper limb	Reference	Reference	Reference	-
Eyelid	0.96 (0.56-1.65)	0.76 (0.34-1.72)	0.51 (0.27-0.96)	.87
Unknown primary	-	0.65 (0.0.51-0.82)	0.81 (0.54-1.23)	.32

CI, Confidence interval; *HR*, hazard ratio.

with regional and distant metastatic disease, but the variance in outcomes between primary sites appears to decrease as the burden of disease increases, suggesting that the effect of tumor primary site on MCC disease-specific death is less prominent with increasing stage. Exceptions include the eyelid

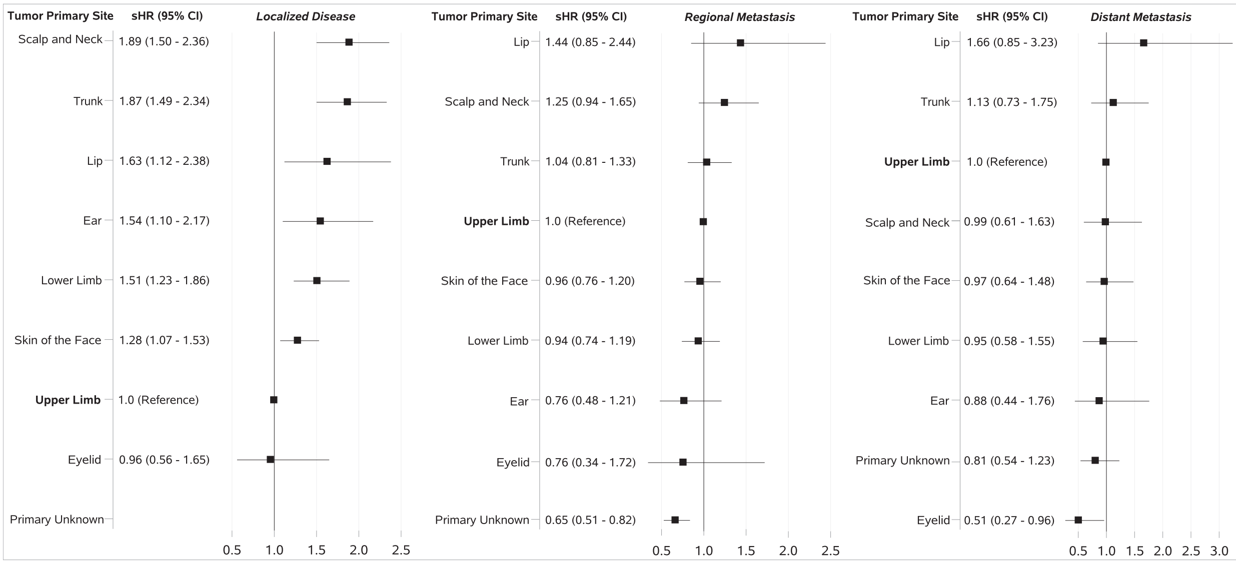

Fig 1. Forest plot of tumor primary site subdistribution hazard ratios for each stage at diagnosis. The sHRs are ordered by magnitude within each stage. Interaction between tumor primary site and stage at diagnosis is demonstrated by the order of the sHRs changing across tumor primary site with increasing stage. *sHRs*, Subdistribution hazard ratios.

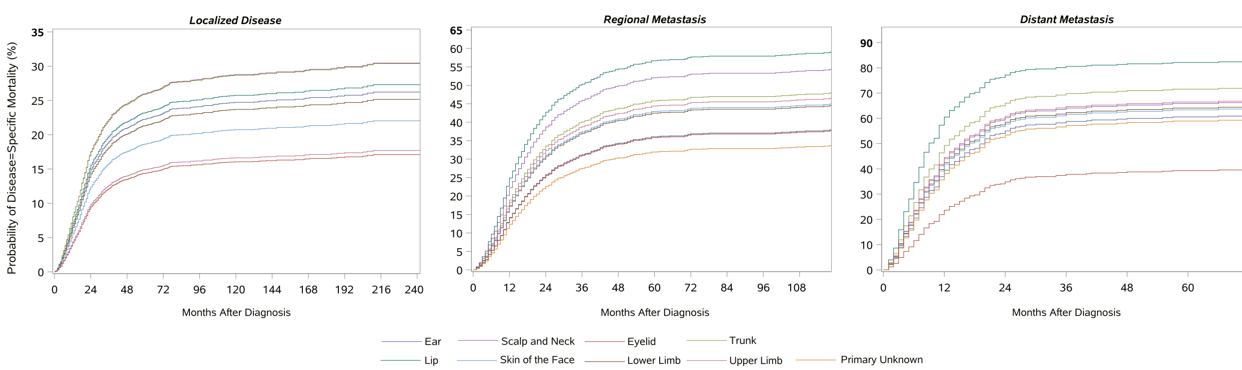

Fig 2. Cumulative Merkel cell carcinoma-specific mortality incidence by tumor primary site, stratified by metastatic progression of disease. Cumulative Merkel cell carcinoma-specific mortality incidence is the outcome measured on each y-axis, representing the probability of Merkel cell carcinoma disease-specific death. Scale of y-axis varies according to stage at diagnosis. Time in months after diagnosis is represented on the x-axis. Scale of x-axis also varies according to stage at diagnosis. Graphs display cumulative Merkel cell carcinoma-specific mortality incidence by tumor primary site for each of the 3 stages at diagnosis (ie, localized disease, regional metastasis, and distant metastasis).

(notably lower increase in disease-specific death) and lip (notably higher increase in disease-specific death). Our results suggest that tumor primary site could be considered when determining MCC prognosis.

Prior studies evaluating tumor primary site as a prognostic factor for MCC are limited. Studies either have included few tumor primary sites or have combined multiple tumor sites encompassing large body regions for comparison. Although Agelli et al[21] noted lower overall survival rates for head/neck MCCs relative to upper limb MCCs, findings from recent studies have been mixed.[5,6] Smith et al[22] demonstrated increased mortality for lip MCC compared to other head/neck primary sites, but this association did not persist in their comparison of all head/neck primary sites to tumor primary sites beyond the head/neck region.

In contrast, we found that tumor primary sites within the head/neck region differ in MCC disease-specific death, both among each other and when compared to other sites across the body. The lip

Table III. Five-year estimated probabilities of Merkel cell carcinoma disease-specific death, by tumor primary site (1973-2016)

	Localized disease (n = 5626)	Regional metastasis (n = 2112)	Distant metastasis (n = 452)
	% (95% CI)	% (95% CI)	% (95% CI)
Tumor primary site			
Scalp/neck	26.0 (23.0-30.4)	52.0 (43.1-60.5)	65.9 (48.6-77.2)
Trunk	25.9 (22.9-30.8)	45.8 (39.1-51.5)	71.6 (55.3-78.6)
Lip	23.2 (17.2-32.1)	56.7 (40.1-77.4)	82.1 (62.0-98.7)
Ear	22.3 (17.4-28.7)	36.0 (24.6-51.2)	60.5 (38.4-84.3)
Lower limb	21.3 (19.0-25.3)	42.4 (36.3-47.8)	64.0 (47.2-75.1)
Face (other)	18.6 (16.8-21.3)	42.9 (36.5-48.8)	63.4 (52.0-70.6)
Upper limb	14.9 (13.4-17.1)	44.4 (39.7-49.2)	66.5 (50.3-75.3)
Eyelid	14.3 (9.0-23.3)	34.8 (18.8-67.9)	39.3 (25.7-57.8)
Unknown primary	-	31.9 (26.5-36.0)	58.9 (36.7-57.9)

CI, Confidence interval; *HR*, hazard ratio; *n*, number of patients.

primary site specifically has the highest probability of disease-specific death among MCC with regional and distant metastasis, in addition to being among the highest for localized disease. However, ear and eyelid tumors had among the lowest probabilities of MCC disease-specific death. Thus, grouping head/neck tumor primary sites with heterogenous outcomes (e.g., lip and ear/eyelid) in analyses[12] may lead to biased mortality estimates for MCC.

Despite similar anatomic locations, there are potential explanations for differences in MCC disease-specific death among head/neck tumors. Tumor primary site has been studied as a prognostic factor in cutaneous melanoma. Melanoma-specific mortality has similarly been shown to occur at higher rates for head/neck tumors, which is believed to be a result of a higher tendency for missed lymphatic metastases due to the complex lymphatic drainage in this region.[23,24] Our observation of differences in CMMI among different head/neck primary sites is inadequately explained by this theory, however. Perhaps in MCC, certain head/neck sites carry a higher propensity for micrometastatic disease, making these tumors more difficult to identify clinically. Before the adoption of routine sentinel lymph node biopsy, these micrometastatic tumors may have been less likely to receive lymphadenectomy, thus leading to worse CMMI. In a post-hoc sensitivity analysis, however, we found tumor primary site to remain predictive of differences in 5-year CMMI between eras of MCC treatment.

Harms et al[25] recently reported an association between tumor primary site and Merkel cell polyomavirus status. Specifically, virus-positive MCCs tended to appear on the extremities and carry fewer mutations, whereas virus-negative MCCs tended to appear in the head/neck region and carry more mutations. Notably, Harms et al[25] demonstrate that virus-positive MCCs had better disease-specific survival (Hazard ratio 0.34). Thus, the impact of tumor primary site on MCC prognosis may be mediated by differences in Merkel cell polyomavirus status. Perhaps, then, the effect modification demonstrated in our analysis is mediated by a propensity for virus-positive MCCs to remain localized for longer periods of time, with fewer mutations that promote disease progression.

We observed lower CMMI for MCC of unknown primary site compared to most, but not all, cutaneous primary sites. Previous literature has suggested that MCC of unknown primary is associated with better prognosis across various outcome measures, including improved overall survival and decreased risk of recurrence,[10] decreased rate of disease-specific death,[26] and decreased rate of visceral metastasis and all-cause mortality.[5]

It remains unclear whether MCC of unknown primary carries a survival benefit at the distant metastatic stage.[26] Our findings suggest that, while MCC of unknown primary retains a survival advantage compared to cutaneous tumor primary sites, this advantage is reduced at the distant metastatic stage. These results do not exclude the possibility of heterogenous subtypes within MCC of unknown primary. Vandeven et al[26] demonstrated that MCC of unknown primary located within skin-draining lymph nodes may have superior outcomes compared to MCC of unknown primary located within non-skin-draining lymph nodes. We were unable to account for such differences within our analysis.

Clinicians might consider incorporating tumor primary site into future MCC treatment guidelines to a greater extent. Current indications for clinical decision-making minimally involve tumor primary site, with the exception of rare cases that involve deciding between radiation therapy and surgery.[27] To our knowledge, the current literature does not describe differences in therapeutic outcomes between MCC tumor primary sites, beyond those observed in case reports or series (eg, periocular Mohs micrographic surgery for local control and tissue preservation).[28] Our findings invite consideration of escalating primary therapy or adjusting routine follow up after primary therapy for MCCs presenting at tumor primary sites that have a higher incidence of disease-specific death. We encourage future studies to address these and related questions.

Our study uses Fine and Gray's proportional subdistribution hazards regression to account for competing risks. The statistical literature shows that the Kaplan-Meier method, which has traditionally been used in studies demonstrating MCC survival analyses, can overestimate mortality and bias outcomes when competing risks are present, especially when the competing risk is common.[18,29,30] For dermatologic malignancies, such as MCC, occurring mostly in a population of advanced age,[1] death from other causes not related to MCC (ie, the competing risk) occurs frequently (35.2% of deaths in our study). As such, estimating mortality outcomes using the traditional Kaplan-Meier approach is likely to introduce bias.[30] Fine and Gray's method is particularly apt for testing the significance of the effect of a covariate (eg, tumor primary site) on the cumulative incidence of an outcome (eg, disease-specific death).

Limitations in our study largely relate to weaknesses inherent in SEER. First, data related to tumor size were missing for a substantial proportion of patients, which precluded stratification by stage according to current American Joint Committee on Cancer guidelines.[3] Second, tumor primary site designation in SEER is based on pathology and/or operative reports; therefore, it is plausible that when a tumor primary site cannot be determined by these reports, it is erroneously coded as an unknown primary. We attempted to minimize this limitation by analyzing the cohort for inconsistencies with other tumor variables and excluding any incongruent cases (eg, patient for whom tumor size is listed, despite tumor primary site being categorized as an unknown primary). Third, certain prognostic factors for MCC (eg, immunosuppression or histopathologic features) are unavailable within SEER and therefore could not be included in our analysis. Fourth, and finally, patients diagnosed prior to the widespread adoption of sentinel lymph node biopsy may have harbored micrometastatic disease, but would have been erroneously staged in SEER as having localized MCC at diagnosis.

CONCLUSION

This study investigated the effect of tumor primary site on the incidence of MCC disease-specific death using a large national cancer database. We demonstrate that tumor primary site impacts the probability of MCC disease-specific death. This finding varied by stage at diagnosis, suggesting that tumor primary site may be a useful prognostic indicator for MCC. Our findings should encourage future research on care pathways that account for the differential mortality risk based on tumor primary site which MCC patients might face after primary treatment.

Conflicts of interest

None disclosed.

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